Co-optimization of Bidirectional Meeting Point Assignment and Dynamic Routing for Many-to-Many Demand-Responsive Transit with Passenger Choice

An Jin[a], Tianliang Liu[a,*]

[a]*School of Transportation Science and Engineering, Beihang University, Beijing 100191, China*

Abstract

Demand-responsive transit (DRT) offers flexible mobility in low-density urban areas, but door-to-door service is operationally costly and difficult to scale. Meeting-point-based DRT reduces vehicle detours by consolidating pickups and dropoffs at pre-defined locations, yet existing work assigns meeting points only on the pickup side and ignores passenger heterogeneity in mode choice. We address this gap by proposing a many-to-many DRT model with bidirectional meeting point sets (M_r^P for pickup, M_r^D for dropoff) and explicit Multinomial Logit (MNL) passenger choice with three passenger types and an outside option. The model is formulated as a coupled three-layer system — service offer generation, passenger response, and dynamic dispatch — and solved with a Mixed-Integer Linear Program (MILP) exact benchmark for small instances and a rolling horizon Adaptive Large Neighbourhood Search (ALNS) heuristic for large-scale deployment. Numerical experiments on synthetic scenarios calibrated to Chinese suburban conditions show that, at equal served share (≈22.8%), the full bidirectional model achieves

*Corresponding author

11.1 vkm/trip compared to **17.1 vkm/trip** for the DoorToDoor (capped) baseline — a **35.0%** improvement in per-trip vehicle utilisation that cannot be attributed to coverage differences (endogenous matched-coverage comparison). Under the unconstrained comparison (FullModel 20.8% vs. DoorToDoor 61.0% served share), FullModel achieves 15.1 vkm/trip versus 21.3 vkm/trip (29.2% improvement). For reference, a post-hoc matched-coverage estimate yields 10.9 vs. 42.3 vkm/trip (74.3% improvement), providing a conservative lower bound. Equity analysis across passenger types reveals a Gini coefficient of 0.1216 in acceptance rates: time-sensitive passengers are systematically disadvantaged (14.4% acceptance) relative to price-sensitive (25.4%) and walk-sensitive (20.2%) passengers. Based on these findings, we derive five actionable policy recommendations for Chinese city DRT operators, including a walking radius threshold of 1000 m for viable bidirectional deployment and a minimum fleet ratio of 15 vehicles per 100 daily requests.
Keywords: demand-responsive transit, meeting points, passenger choice, ALNS, rolling horizon, equity

1. Introduction

Demand-responsive transit (DRT) has emerged as a promising solution for providing flexible, on-demand mobility in low-density urban areas where fixed-route bus services are economically unviable. In China, rapid suburbanisation has created large residential zones with sparse transit coverage, making DRT an increasingly important policy instrument for urban planners and transport authorities. However, the dominant door-to-door DRT model — in which vehicles travel directly between each passenger's origin and destination — is operationally costly: vehicles accumulate large detour distances serving spatially dispersed requests, and fleet utilisation remains low. Meeting-point-based DRT addresses this inefficiency by requiring passengers to walk to and from pre-defined consolidation points (e.g., bus stops, commercial landmarks), enabling vehicles to serve multiple passengers along shared route segments. The trade-off is clear: meeting points reduce vehicle-kilometres but impose a walking burden on passengers, and passengers who find the required walk unacceptable will reject the service offer and use an alternative mode. Jointly optimising the spatial assignment of meeting points and the vehicle routing schedule — while explicitly accounting for passenger heterogeneity in mode choice — is therefore the central operational challenge for next-generation DRT systems.

Existing literature on meeting-point DRT has made important progress but leaves a critical gap. The Dial-a-Ride Problem (DARP) literature provides a rich foundation for vehicle routing with time windows and capacity constraints [1, 2]. Recent extensions have introduced meeting points into the DARP framework. Cortenbach et al. [3] proposed the DARPmp, the first formulation of DARP with meeting points, using a Tabu Search heuristic to solve instances with single-sided meeting point assignment: pickup locations are flexible (passengers walk to a nearby stop), but dropoff locations remain fixed at the passenger's destination. This single-sided approach captures part of the efficiency gain but leaves the dropoff leg unoptimised. Crucially, Cortenbach et al. [3] do not model passenger choice: all offered meeting points are assumed to be accepted, ignoring the reality that passengers with low walking tolerance will reject inconvenient offers. On the dynamic side, Wu et al. [4] proposed a rolling horizon framework for online DRT re-optimisation, demonstrating that periodic re-assignment of requests to vehicles significantly reduces operational cost under dynamic demand. However, Wu et al. [4] do not incorporate meeting points: their model retains door-to-door service. No existing work combines: (a) bidirectional meeting point sets (M_r^P for pickup, M_r^D for dropoff), (b) explicit Multinomial Logit (MNL) passenger choice with heterogeneous passenger types and an outside option, and (c) rolling horizon re-optimisation in a many-to-many DRT setting. This is the gap the present paper fills.

We propose a many-to-many DRT model with bidirectional meeting point sets. For each request r, the system constructs a candidate pickup set $M_r^P \subseteq \mathcal{M}$ (meeting points within walking radius ρ^P of the passenger's origin) and a candidate dropoff set $M_r^D \subseteq \mathcal{M}$ (within ρ^D of the destination). The system jointly optimises the selection of pickup meeting point $m_r^P \in M_r^P$ and dropoff meeting point $m_r^D \in M_r^D$ for each accepted request, together with vehicle routing and scheduling. Passenger response is modelled via a Multinomial Logit (MNL) model with three passenger types — price-sensitive, time-sensitive, and walk-sensitive — each with distinct utility coefficients over walking time, waiting time, in-vehicle travel time, and fare. An outside option allows passengers to reject all offered bundles and use an alternative mode. The full model is formulated as a coupled three-layer system: (1) service offer generation, which enumerates feasible bundles $\mathcal{B}_r = M_r^P \times M_r^D$ and computes MNL choice probabilities; (2) passenger response, which determines the accepted bundle based on MNL utilities; and (3) dynamic dispatch, which routes vehicles using a rolling horizon ALNS heuristic with re-optimisation at fixed time intervals. We solve small instances exactly using a Mixed-Integer Linear Program (MILP) implemented in Gurobi, and large instances with the rolling horizon ALNS heuristic. Numerical experiments on synthetic scenarios calibrated to Chinese suburban conditions validate the model and quantify the efficiency and equity implications of bidirectional meeting point assignment.

The contributions of this paper are as follows:

1. First formulation of many-to-many DRT with bidirectional meeting point sets (M_r^P, M_r^D) and MNL passenger choice as a coupled three-layer model.
2. MILP exact benchmark and rolling horizon ALNS heuristic for the bidirectional DARP with passenger choice.
3. Empirical demonstration of a **35.0%** vehicle efficiency gain in per-trip vehicle utilisation at equal served share (FullModel 11.1 vs. DoorToDoor (capped) 17.1 vkm/trip, endogenous matched-coverage comparison), and a 29.2% gain under the unconstrained comparison (15.1 vs. 21.3 vkm/trip).
4. Equity analysis across three passenger types (Gini coefficient = 0.1216) and five actionable policy recommendations for Chinese city DRT operators.

The remainder of this paper is organised as follows. Section 2 reviews related work on DARP, meeting-point DRT, and passenger choice modelling. Section 3 presents the problem formulation, including the three-layer coupled model and the MNL choice structure. Section 6 describes the solution methodology: the MILP exact benchmark and the rolling horizon ALNS heuristic. Section 7 reports numerical experiments, including ablation studies and sensitivity analysis across demand density, walking tolerance, and fleet size. Section 8 discusses policy implications for Chinese city DRT operators. Section 9 concludes and outlines directions for future research.

2. Literature Review

2.1. Dial-a-Ride and Demand-Responsive Transit

The Dial-a-Ride Problem (DARP) is a variant of the vehicle routing problem with time windows in which a fleet of vehicles must transport passengers between individual origin–destination pairs while respecting time window constraints, maximum ride time limits, and vehicle capacity [1]. Since its formal introduction, the DARP has attracted sustained attention as a model for paratransit, elderly and disabled transport, and, more recently, on-demand mobility services in low-density urban areas [5, 6].

Demand-responsive transit (DRT) extends the DARP framework to a broader operational context in which service is triggered by real-time passenger requests rather than fixed schedules. The growth of smartphone-based booking platforms and GPS-enabled fleets has made DRT increasingly viable as a first- and last-mile solution in suburban and peri-urban areas where fixed-route bus services are economically unsustainable [7]. Comprehensive surveys by Ho et al. [2] and Pillac et al. [8] document the rapid expansion of DRT research, covering exact methods, metaheuristics, and online algorithms for both static and dynamic problem variants.

A defining characteristic of classical DARP formulations is the assumption of door-to-door service: vehicles travel directly to each passenger's origin and destination address. While this maximises passenger convenience, it generates highly fragmented routes with significant detour and deadhead mileage, particularly when demand is spatially dispersed [9]. The present paper relaxes the door-to-door assumption by introducing bidirectional meeting point sets, allowing both pickup and dropoff locations to be selected from a pre-defined network of accessible stops. This structural change is the central departure from classical DARP and motivates the literature reviewed in the following subsection.

2.2. Meeting Points and Virtual Stops

The concept of meeting points — fixed or semi-fixed locations where passengers walk a short distance to board or alight — was introduced into the ridesharing literature by Stiglic et al. [10], who demonstrated that allowing passengers to walk to nearby virtual stops substantially reduces vehicle detour and increases the number of feasible matches in a shared-ride system. A follow-up study [11] quantified the trade-off between walking burden and system efficiency, showing that even modest walking tolerances (200–500 metres) can yield significant reductions in total vehicle kilometres travelled without materially degrading passenger experience.

Building on this foundation, Cortenbach et al. [3] introduced the Dial-a-Ride Problem with Meeting Points (DARPmp), the first formal DARP extension to incorporate meeting point selection as a decision variable. Their formulation allows the pickup location to be chosen from a candidate set while the dropoff location remains fixed at the passenger's destination address — a *single-sided* meeting point design. The problem is solved with a Tabu Search metaheuristic applied to a static batch of requests. Cortenbach et al.

[3] report substantial efficiency gains over door-to-door baselines, confirming the operational value of flexible pickup locations in many-to-many DRT.

However, the DARPmp of Cortenbach et al. [3] has two important limitations that the present paper addresses. First, the single-sided design leaves dropoff flexibility unexploited: passengers must still be delivered to their exact destination, which constrains route consolidation on the egress leg. Second, the model treats passenger acceptance as deterministic — once the system assigns a meeting point, the passenger is assumed to comply regardless of walking distance or service quality. In practice, passengers exercise choice, and a model that ignores this risks over-optimistic efficiency estimates. The present paper extends DARPmp to a *bidirectional* design in which both the pickup meeting point $m_r^P \in M_r^P$ and the dropoff meeting point $m_r^D \in M_r^D$ are decision variables, and couples this with an explicit multinomial logit (MNL) passenger choice model.

Fielbaum et al. [12] studied bidirectional walking flexibility in ridepooling, showing that allowing passengers to walk to both pickup and dropoff points reduces vehicle detours and improves system efficiency. Their analysis demonstrates that the efficiency gains from dropoff-side flexibility are comparable in magnitude to those from pickup-side flexibility, providing empirical motivation for the bidirectional design adopted in the present paper. Our work extends this insight to the many-to-many DRT setting with explicit MNL passenger choice and online re-optimisation: unlike Fielbaum et al. [12], who study a static ridepooling context, we address the online assignment problem where meeting points must be selected in real time as requests arrive, and passenger acceptance is modelled probabilistically rather than assumed.

2.3. Passenger Choice in Demand-Responsive Transit

Discrete choice modelling provides the theoretical foundation for representing heterogeneous passenger preferences in transportation systems. The multinomial logit (MNL) model, derived from random utility maximisation [13], is the workhorse of mode choice analysis and has been applied extensively to transit demand, ridesharing adoption, and on-demand mobility [14]. In the MNL framework, each passenger selects the alternative that maximises their perceived utility, with choice probabilities determined by the softmax function over deterministic utility components.

The application of discrete choice models to DRT optimisation is relatively recent. Most DRT scheduling papers treat passenger acceptance as a binary constraint: a request is either served within its time window or rejected, with no intermediate probability of acceptance. This deterministic treatment ignores the empirical reality that passengers differ in their sensitivity to walking distance, waiting time, in-vehicle travel time, and fare [14]. Heterogeneous preferences are particularly consequential in meeting-point DRT, where the system must select not only *whether* to serve a request but *where* to pick up and drop off the passenger — a choice that directly affects the utility experienced.

The present paper builds on the MNL framework developed in Jin and Liu [15] for a many-to-one DRT setting with dynamic pricing. That work demonstrated that explicitly modelling passenger type heterogeneity (price-sensitive, time-sensitive, and walk-sensitive segments) yields more accurate demand forecasts and better-calibrated routing decisions than deterministic acceptance models. The present paper extends this framework to the many-to-many bidirectional meeting-point setting, retaining the three-type MNL structure and the outside option (passengers may reject all offered bundles) as core modelling elements. The outside option is critical: it ensures that the optimisation does not assign meeting points so inconvenient that passengers would rationally decline service, a failure mode absent from deterministic DARP formulations.

2.4. Dynamic Scheduling and Rolling Horizon Methods

Real-world DRT operates in an online environment: requests arrive continuously, vehicles are in motion, and schedules must be updated in near-real-time. The online DARP [5] and its dynamic variants have been studied extensively, with solution approaches ranging from greedy insertion heuristics to population-based metaheuristics and reinforcement learning [8, 2].

Rolling horizon (receding horizon) methods decompose the infinite-horizon online problem into a sequence of finite-horizon subproblems, each solved to optimality or near-optimality before the horizon advances [16]. Bent and Van Hentenryck [16] established the theoretical basis for rolling horizon approaches in vehicle routing, showing that re-optimisation over a sliding window can closely approximate clairvoyant offline solutions when the window length is calibrated to the request arrival rate. This insight motivates the rolling horizon architecture adopted in the present paper.

The Adaptive Large Neighbourhood Search (ALNS) of Ropke and Pisinger [17] is the dominant metaheuristic for DARP and related problems, combining destroy-and-repair operators with an adaptive weight mechanism that favours operators performing well on recent iterations. ALNS has been applied to static DARP [6], dynamic ridesharing [9], and, most recently, to online DRT re-optimisation.

Wu et al. [4] present a rolling horizon ALNS framework for many-to-many DRT that re-optimises vehicle routes at fixed time intervals as new requests arrive. Their computational experiments on synthetic and real-world instances demonstrate that rolling horizon ALNS achieves near-optimal acceptance rates with tractable computation times. However, the formulation of Wu et al. [4] assumes door-to-door service — no meeting points are considered — and treats passenger acceptance as deterministic. The present paper integrates the rolling horizon ALNS architecture of Wu et al. [4] with the bidirectional meeting point assignment and MNL choice model described above, yielding a unified online co-optimisation framework.

2.5. Research Gap and Positioning

Table 1 summarises the positioning of this paper relative to the two most closely related prior works. No existing study simultaneously addresses bidirectional meeting point sets (M_r^P and M_r^D), stochastic passenger choice via MNL with an outside option, rolling horizon re-optimisation, and a many-to-many service structure. This paper is the first to jointly address all four dimensions.

3. Problem Formulation

3.1. Network and Meeting Points

We model the service area as a directed graph $G = (N, A)$, where N is the set of nodes comprising passenger origins, destinations, and meeting points,

Table 1: Positioning relative to prior work

Study	Bidirectional MPs	Passenger Choice	Rolling Horizon	Many-to-Ma
Cortenbach et al. (2024)	No (single-sided)	No	No	Yes
Wu et al. (2025)	No	No	Yes	Yes
This paper	Yes	Yes (MNL)	Yes	Yes

and A is the set of directed arcs. Travel time between any two locations $i, j \in N$ is given by $t(i, j) = d(i, j)/\bar{v}$, where $d(i, j)$ denotes Euclidean distance and $\bar{v}$ is the average vehicle speed.

A finite set $\mathcal{M}$ of pre-defined fixed meeting points (e.g., bus stops, commercial landmarks) is given. For each request r, the candidate pickup set $M_r^P \subseteq \mathcal{M}$ is the subset of meeting points within walking radius ρ^P of the passenger's origin o_r, and the candidate dropoff set $M_r^D \subseteq \mathcal{M}$ is the subset within walking radius ρ^D of the destination δ_r. Formally:

$$M_r^P = \{m \in \mathcal{M} \mid d(o_r, m) \leq \rho^P\}, \qquad M_r^D = \{m \in \mathcal{M} \mid d(\delta_r, m) \leq \rho^D\}. \tag{1}$$

Each request $r \in \mathcal{R}$ is characterized by a tuple $(o_r, \delta_r, e_r, l_r, k_r)$, where $[e_r, l_r]$ is the pickup time window and $k_r \in \mathcal{K}$ denotes the passenger type. Requests arrive online over a continuous planning horizon. The fleet $\mathcal{V}$ consists of vehicles with heterogeneous capacities Q_v and maximum shift durations $T_v^{\max}$.

3.2. Service Offer Bundle

For each request r, a *service offer bundle* is a tuple

$$b = (m_r^P,\ m_r^D,\ \tau_r,\ p_r),$$

where:

- $m_r^P \in M_r^P$ is the assigned pickup meeting point,
- $m_r^D \in M_r^D$ is the assigned dropoff meeting point,
- $\tau_r \in [e_r, l_r]$ is the scheduled pickup time at m_r^P,
- $p_r \geq 0$ is the fare charged to the passenger (fixed in this model).

The set of all feasible bundles for request r is $\mathcal{B}_r = M_r^P \times M_r^D$ (with τ_r and p_r treated as service parameters determined by the system). The service offer bundle b is the central decision object: the system selects one bundle from $\mathcal{B}_r$ or rejects the request.

The bidirectional meeting-point structure distinguishes this model from prior work. In many-to-one DRT, only the pickup side is flexible; here both pickup and dropoff meeting points are co-optimized, creating a richer trade-off between vehicle routing efficiency and passenger walking burden.

3.3. System Objective

The system minimizes a weighted sum of operational and passenger service costs over the planning horizon:

$$\min \quad \alpha_1 C^{op} + \alpha_2 C^{wait} + \alpha_3 C^{walk} + \alpha_4 C^{IVT} + \alpha_5 C^{rej}, \tag{2}$$

where:

$$C^{op} = \sum_{v \in \mathcal{V}} c_v \cdot D_v, \tag{3}$$

$$C^{wait} = \sum_{r \in \mathcal{R}} z_r \cdot (\tau_r - t_r^{\text{req}}), \tag{4}$$

$$C^{walk} = \sum_{r \in \mathcal{R}} z_r \cdot \big(d(o_r, m_r^P) + d(m_r^D, \delta_r)\big), \tag{5}$$

$$C^{IVT} = \sum_{r \in \mathcal{R}} z_r \cdot \text{IVT}_{rb}, \tag{6}$$

$$C^{rej} = \sum_{r \in \mathcal{R}} (1 - z_r) \cdot \Gamma. \tag{7}$$

Here D_v is the total distance driven by vehicle v, c_v is cost per unit distance, t_r^{req} is the time request r was submitted, and $\Gamma > 0$ is the rejection penalty. The weights $\alpha_1, \ldots, \alpha_5 \geq 0$ allow the operator to balance vehicle efficiency against service quality and equity.

3.4. Operational Constraints

The feasibility region is defined by the following fourteen constraints.

Vehicle Capacity..

$$q_v(i) \leq Q_v \quad \forall v \in \mathcal{V},\ \forall i \in \sigma_v \tag{8}$$

Time Windows..

$$s_{v_r, m_r^P} \geq e_r \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{9}$$

$$s_{v_r, m_r^P} \leq l_r \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{10}$$

In-Vehicle Ride Time..

$$s_{v_r, m_r^D} - s_{v_r, m_r^P} \leq L^{\max} \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{11}$$

Walking Radius..

$$d(o_r,\, m_r^P) \leq \rho^P \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{12}$$

$$d(\delta_r,\, m_r^D) \leq \rho^D \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{13}$$

Precedence: Pickup Before Dropoff..

$$\pi_r^P < \pi_r^D \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{14}$$

$$s_{v_r,\, m_r^P} + t(m_r^P,\, m_r^D) \leq s_{v_r,\, m_r^D} \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{15}$$

Route Time Consistency..

$$s_{v,\, i+1} \geq s_{v,\, i} + t(\sigma_v(i),\, \sigma_v(i+1)) \quad \forall v \in \mathcal{V},\ \forall i < |\sigma_v| \tag{16}$$

Maximum Route Duration..

$$s_{v,\, |\sigma_v|} - s_{v,\, 1} \leq T_v^{\max} \quad \forall v \in \mathcal{V} \tag{17}$$

Domain Constraints..

$$z_r \in \{0, 1\} \qquad \forall r \in \mathcal{R} \tag{18}$$

$$v_r \in \mathcal{V} \qquad \forall r \in \mathcal{R},\ z_r = 1 \tag{19}$$

$$m_r^P \in M_r^P \qquad \forall r \in \mathcal{R},\ z_r = 1 \tag{20}$$

$$m_r^D \in M_r^D \qquad \forall r \in \mathcal{R},\ z_r = 1 \tag{21}$$

Table 2 summarizes all fourteen constraints.

3.5. Online Decision Vector

Upon arrival of request r, the system must immediately determine the *online decision vector*:

$$\mathbf{x}_r = (z_r,\ m_r^P,\ m_r^D,\ v_r,\ \pi_r^P,\ \pi_r^D), \tag{22}$$

Table 2: Summary of operational constraints

Label	**Class**	**Variables Involved**
8	Vehicle Capacity	$q_v(i)$, Q_v, σ_v
9	Time Windows (early)	s_{v_r,m_r^P}, e_r
10	Time Windows (late)	s_{v_r,m_r^P}, l_r
11	In-Vehicle Ride Time	s_{v_r,m_r^D}, s_{v_r,m_r^P}, $L^{\max}$
12	Walking Radius (pickup)	$d(o_r, m_r^P)$, ρ^P
13	Walking Radius (dropoff)	$d(\delta_r, m_r^D)$, ρ^D
14	Precedence (position)	π_r^P, π_r^D
15	Precedence (time)	s_{v_r,m_r^P}, s_{v_r,m_r^D}, $t(m_r^P, m_r^D)$
16	Route Time Consistency	$s_{v,i}$, $s_{v,i+1}$, $t(\sigma_v(i), \sigma_v(i+1))$
17	Maximum Route Duration	$s_{v,1}$, $s_{v,\lvert\sigma_v\rvert}$, $T_v^{\max}$
18	Domain (acceptance)	z_r
19	Domain (vehicle)	v_r
20	Domain (pickup point)	m_r^P, M_r^P
21	Domain (dropoff point)	m_r^D, M_r^D

where $z_r \in \{0, 1\}$ is the acceptance indicator, $m_r^P \in M_r^P$ and $m_r^D \in M_r^D$ are the assigned meeting points, $v_r \in \mathcal{V}$ is the assigned vehicle, and $\pi_r^P < \pi_r^D$ are the insertion positions of the pickup and dropoff nodes in vehicle v_r's route. The decision must be made online — before future requests are known — and subject to all feasibility constraints above.

4. Passenger Choice Model

4.1. MNL Utility Function

We model passenger choice using a Multinomial Logit (MNL) model. For request r of passenger type $k \in \mathcal{K}$, the deterministic utility of service offer bundle $b = (m_r^P, m_r^D, \tau_r, p_r)$ is:

$$U_{rb}^k = \beta_1^k \cdot \text{Walk}_{rb} + \beta_2^k \cdot \text{Wait}_{rb} + \beta_3^k \cdot \text{IVT}_{rb} + \beta_4^k \cdot p_r \tag{23}$$

where:

- $\text{Walk}_{rb} = d(o_r, m_r^P) + d(m_r^D, \delta_r)$ is the total walking distance (pickup walk plus dropoff walk) in meters,
- $\text{Wait}_{rb} = \tau_r - t_r^{\text{req}}$ is the waiting time from request submission to scheduled pickup (minutes),
- $\text{IVT}_{rb} = s_{v_r, m_r^D} - s_{v_r, m_r^P}$ is the in-vehicle travel time from pickup to dropoff meeting point (minutes),
- $p_r \geq 0$ is the fare (CNY),
- $\beta_1^k, \beta_2^k, \beta_3^k \leq 0$ are disutility coefficients for walking, waiting, and in-vehicle time (negative: more time/distance reduces utility),
- $\beta_4^k \leq 0$ is the price disutility coefficient.

The random utility is $U_{rb}^k + \varepsilon_{rb}$ where ε_{rb} is i.i.d. Gumbel$(0, 1)$, yielding the standard MNL form. The utility function captures the bidirectional walking cost explicitly through Walk_{rb}, which depends on both m_r^P and m_r^D — a key distinction from one-sided meeting-point models.

4.2. *Single-Offer Mechanism and Binary Logit Acceptance*

The system operates a *single-offer mechanism*: Layer 1 selects the best feasible bundle $b^* \in \mathcal{B}_r$ and presents *only* b^* to the passenger. The passenger never observes the full set $\mathcal{B}_r$; they choose between accepting b^* or taking the outside option (e.g., private car, walking, trip cancellation).

The utility of the outside option is normalized to zero:

$$U_{r0}^k = 0 \tag{24}$$

Because the passenger's choice set contains exactly two alternatives — b^* and the outside option — the acceptance probability follows a *binary logit* model:

$$P_{\text{accept}}(b^*) = \frac{\exp(U_{rb^*}^k)}{\exp(U_{r0}^k) + \exp(U_{rb^*}^k)} = \frac{\exp(U_{rb^*}^k)}{1 + \exp(U_{rb^*}^k)} \tag{25}$$

The realized acceptance indicator is:

$$z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*)) \tag{26}$$

This formulation is behaviorally consistent with the single-offer mechanism: the passenger responds to the one bundle they are shown, not to a hypothetical full menu. The binary logit is the special case of MNL with $|\text{choice set}| = 2$, so the standard random-utility interpretation (i.i.d. $\text{Gumbel}(0, 1)$ errors) is preserved.

4.3. *Passenger Heterogeneity*

We distinguish three passenger types $\mathcal{K} = \{\text{price}, \text{time}, \text{walk}\}$ with different sensitivity profiles. Table 3 summarizes the qualitative ordering of coefficients.

Table 3: Passenger type coefficient profiles (qualitative ordering)

Type	β_1^k (Walk)	β_2^k (Wait)	β_3^k (IVT)	β_4^k (P
Price-sensitive	moderate	moderate	moderate	high (most
Time-sensitive	moderate	high (most negative)	high (most negative)	lo
Walk-sensitive	high (most negative)	moderate	moderate	mode

Formally, the ordering constraints are:

$$|\beta_4^{\text{price}}| > |\beta_4^{\text{time}}|, \quad |\beta_4^{\text{price}}| > |\beta_4^{\text{walk}}| \tag{27}$$

$$|\beta_2^{\text{time}}| + |\beta_3^{\text{time}}| > |\beta_2^{\text{price}}| + |\beta_3^{\text{price}}| \tag{28}$$

$$|\beta_1^{\text{walk}}| > |\beta_1^{\text{price}}|, \quad |\beta_1^{\text{walk}}| > |\beta_1^{\text{time}}| \tag{29}$$

Baseline parameter values for simulation:

$$(\beta_1^{\text{price}}, \beta_2^{\text{price}}, \beta_3^{\text{price}}, \beta_4^{\text{price}}) = (-0.005,\ -0.04,\ -0.02,\ -0.15) \tag{30}$$

$$(\beta_1^{\text{time}}, \beta_2^{\text{time}}, \beta_3^{\text{time}}, \beta_4^{\text{time}}) = (-0.005,\ -0.10,\ -0.08,\ -0.03) \tag{31}$$

$$(\beta_1^{\text{walk}}, \beta_2^{\text{walk}}, \beta_3^{\text{walk}}, \beta_4^{\text{walk}}) = (-0.020,\ -0.04,\ -0.02,\ -0.05) \tag{32}$$

Units: β_1 in utility/meter, β_2 and β_3 in utility/minute, β_4 in utility/CNY. The heterogeneous type structure allows the model to capture the distributional equity effects of meeting-point assignment: walk-sensitive passengers (e.g., elderly, mobility-impaired) are disproportionately affected by distant meeting points, a key policy consideration for DRT design in low-density urban areas.[1]

[1] **Parameter plausibility.** The implied value of time (VOT) for the walk-sensitive type — the most policy-relevant comparator for bidirectional DRT users — is $\text{VOT}_{\text{walk}} =$

Worked utility example.. Consider a walk-sensitive passenger (k = walk) with $\beta_1^{\text{walk}} = -0.020\,\text{utils/m}$, $\beta_2^{\text{walk}} = -0.04\,\text{utils/min}$, $\beta_3^{\text{walk}} = -0.02\,\text{utils/min}$, and $\beta_4^{\text{walk}} = -0.05\,\text{utils/CNY}$. Suppose the system offers a bundle with walking distance $d_{\text{walk}} = 300\,\text{m}$, waiting time $t_{\text{wait}} = 5\,\text{min}$, in-vehicle time $t_{\text{ivt}} = 20\,\text{min}$, and fare $p = 8\,\text{CNY}$. The deterministic utility is:

$$\begin{aligned} U &= \beta_1 \cdot d_{\text{walk}} + \beta_2 \cdot t_{\text{wait}} + \beta_3 \cdot t_{\text{ivt}} + \beta_4 \cdot p \\ &= (-0.020)(300) + (-0.04)(5) + (-0.02)(20) + (-0.05)(8) \\ &= -6.0 - 0.2 - 0.4 - 0.4 = -7.0 \text{ utils.} \end{aligned}$$

With an outside option penalty of $\mu_0 = 5.0$, the acceptance probability is:

$$P_{\text{accept}} = \frac{e^{-7.0}}{e^{-7.0} + e^{-5.0}} = \frac{0.000912}{0.000912 + 0.006738} \approx 0.119. \tag{33}$$

This low acceptance probability (11.9%) reflects the 300 m walking requirement, which is highly penalised for walk-sensitive passengers. Reducing walking to 100 m would raise U to -3.0 and acceptance to approximately 47%, illustrating the sensitivity of demand to meeting point placement.

$(|\beta_1^{\text{walk}}|/|\beta_4^{\text{walk}}|) \times v_w = (0.020/0.05) \times 80 = 32.0\,\text{CNY/min}$, $\text{VOT}_{\text{wait}} = |\beta_2^{\text{walk}}|/|\beta_4^{\text{walk}}| = 0.04/0.05 = 0.80\,\text{CNY/min}$, and $\text{VOT}_{\text{IVT}} = 0.02/0.05 = 0.40\,\text{CNY/min}$ (walking speed $v_w = 80\,\text{m/min}$ assumed). The wait and IVT VOT values (0.80 and 0.40 CNY/min) fall within the Chinese urban benchmark range of 0.8–1.2 CNY/min (waiting) and 0.2–0.4 CNY/min (IVT) reported by Shao et al. [18] for Beijing commuters and Li et al. [19] for DRT users. The walk VOT (32.0 CNY/min) exceeds the literature range (0.4–0.6 CNY/min); this reflects the unit mismatch between β_1 (utility/meter) and β_4 (utility/CNY): at 80 m/min walking speed, the implied per-minute disutility of walking is high relative to the fare sensitivity, consistent with the walk-sensitive type's defining characteristic. Full VOT mapping across all three passenger types is reported in Table 11 (Section 8).

5. Three-Layer Coupled Model

The many-to-many DRT system with bidirectional meeting points operates through three sequentially coupled layers, executed online upon each request arrival and periodically re-optimized via a rolling horizon mechanism. Figure 2 illustrates the architecture. The three layers are: (1) service offer generation, (2) passenger response, and (3) dynamic dispatch and re-optimization.

5.1. Layer 1: Service Offer Generation

Upon arrival of request r at time t_r^{req}, the system generates a set of candidate service offer bundles $\hat{\mathcal{B}}_r \subseteq \mathcal{B}_r$ as follows:

1. **Candidate pickup set:** Compute $M_r^P = \{m \in \mathcal{M} \mid d(o_r, m) \leq \rho^P\}$. Retain the top-$k^{\text{top}}$ candidates ranked by proximity to o_r.
2. **Candidate dropoff set:** Compute $M_r^D = \{m \in \mathcal{M} \mid d(\delta_r, m) \leq \rho^D\}$. Retain the top-$k^{\text{top}}$ candidates ranked by proximity to δ_r.
3. **Bundle enumeration:** For each pair $(m^P, m^D) \in M_r^P \times M_r^D$, evaluate feasibility against constraints 8–16 for each vehicle $v \in \mathcal{V}$ and each insertion position pair (π^P, π^D).
4. **Bundle selection:** Select the bundle $b^* = \arg\max_{b \in \hat{\mathcal{B}}_r} \mathbb{E}[\text{system benefit} \mid b, \mathbf{x}_r]$ that maximizes expected system objective contribution, subject to feasibility.

The output of Layer 1 is the online decision vector $\mathbf{x}_r$ (Equation 22) and the service offer bundle b^* presented to the passenger.

5.2. Layer 2: Passenger Response

Given the offered bundle b^*, passenger r of type k accepts with probability $P_{\text{accept}}(b^*)$ (Equation 25) and rejects with probability $1 - P_{\text{accept}}(b^*)$. The realized acceptance indicator is:

$$z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*)) \tag{34}$$

If $z_r = 1$, the vehicle route σ_{v_r} is updated by inserting the pickup node m_r^P at position π_r^P and the dropoff node m_r^D at position π_r^D. If $z_r = 0$, the decision vector is discarded and vehicle routes remain unchanged.

The coupling between Layer 1 and Layer 2 is *anticipatory*: the system selects b^* to maximize expected system benefit, accounting for the probability that the passenger will accept. This distinguishes the model from a deterministic assignment approach.

5.3. Layer 3: Dynamic Dispatch and Rolling Horizon Re-optimization

Vehicle dispatch follows the committed route schedules $\{\sigma_v, \mathbf{s}_v\}_{v \in \mathcal{V}}$. Every Δ minutes, a rolling horizon re-optimization is triggered over the active request window $[t^{\text{now}},\ t^{\text{now}} + H]$:

1. **Active set:** Identify $\mathcal{R}^{\text{act}}(t^{\text{now}})$ — all accepted requests not yet completed at time t^{now}.
2. **Re-optimization:** Apply ALNS (Adaptive Large Neighborhood Search) to reassign and reorder requests in $\mathcal{R}^{\text{act}}(t^{\text{now}})$ across vehicles, subject to constraints 8–16.
3. **Commitment:** Commit the updated routes for the next Δ minutes; nodes already served are locked and cannot be reassigned.

The rolling horizon creates a feedback loop from Layer 3 back to Layer 1: updated vehicle schedules after re-optimization change the feasibility and cost of future bundle insertions, influencing the next Layer 1 evaluation.

$$\text{Re-optimize at times } t^{\text{now}} = 0,\ \Delta,\ 2\Delta,\ 3\Delta,\ \ldots \tag{35}$$

Typical parameter values: $H = 30$ minutes (horizon window), $\Delta = 5$ minutes (re-optimization frequency). These are configurable and subject to sensitivity analysis in Phase 4.

5.4. Layer Coupling Summary

Table 4 summarizes the information flows between layers.

Table 4: Three-layer model coupling structure

Layer	Input	Output	Coupling
1: Service Generation	Request r, routes $\{\sigma_v\}$	Bundle b^*, decision $\mathbf{x}_r$	Feeds Layer 2
2: Passenger Response	Bundle b^*, type k	Acceptance z_r	Updates routes if $z_r =$
3: Rolling Horizon	Active set $\mathcal{R}^{\text{act}}$	Updated routes $\{\sigma_v\}$	Feeds back to Layer 1

Table 5 details the event sequence and information flows within a single planning cycle; in particular, it shows that Bernoulli sampling ($z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*))$) occurs exclusively in Layer 2, after bundle b^* has been selected by Layer 1.

The three-layer architecture captures the essential feedback structure of online DRT operations: routing decisions shape the offers available to passengers, passenger choices determine realized demand, and periodic re-optimization adapts routes to the evolving state. The notation table (Ta-

Table 5: Timing and decision sequence in the rolling horizon three-layer model. Each row represents one event within a single planning cycle. Re-optimization triggers fire at $t^{\text{now}} = 0, \Delta, 2\Delta, \ldots$ (Equation 35).

Event	**Layer**	**Decision Variables**	**Information Flow**	**Bernoulli?**
Request r arrives at t_r^{req}	1	$m_r^P,\ m_r^D,\ v_r,\ \pi_r^P,\ \pi_r^D$	Routes $\{\sigma_v\}$ in; bundle b^* out	No
Bundle b^* presented to passenger	2	z_r	b^* in; acceptance z_r out	Yes: $z_r \sim \text{Bern}(P_{\text{acc}})$
If $z_r = 1$: route σ_{v_r} updated	3	(route update)	z_r, b^* in; updated σ_{v_r} out	No
Re-opt trigger at $t = k\Delta$	3	$\{\sigma_v\}$ (all vehicles)	$\mathcal{R}^{\text{act}}$ in; optimised $\{\sigma_v\}$ out	No
Updated routes fed back for next request	1	—	Updated $\{\sigma_v\}$ available to Layer 1	No

ble A.12) in the appendix provides a complete reference for all symbols used in this paper.

6. Solution Methodology

The problem defined in Section 3 is NP-hard (see Section 6.3). We develop two complementary solution approaches: an exact MILP formulation solved by Gurobi for small instances, and a rolling horizon Adaptive Large Neighborhood Search (ALNS) heuristic for real-time operation at scale.

6.1. Exact Algorithm: MILP Formulation

For small instances, we formulate the static batch version of the problem as a Mixed-Integer Linear Program (MILP) to obtain provably optimal solutions. These serve as benchmarks for validating heuristic quality.

Decision Variables.. The MILP uses binary assignment variables $x_{r,m^P,m^D,v} \in \{0,1\}$, equal to 1 if request r is assigned to meeting point pair $(m^P, m^D) \in M_r^P \times M_r^D$ and served by vehicle $v \in \mathcal{V}$. Continuous variables $s_{v,i} \geq 0$ represent the scheduled service time at node i in vehicle v's route. The acceptance indicator is recovered as $z_r = \sum_{m^P,m^D,v} x_{r,m^P,m^D,v}$.

Objective.. The MILP minimizes the same weighted cost (Equation 2), with each cost component linearized in terms of $x_{r,m^P,m^D,v}$ and $s_{v,i}$.

Constraints.. The formulation enforces all fourteen operational constraints (8)–(21). Capacity constraint (8) is linearized using standard load-tracking variables. Time-window constraints (9)–(10), ride-time constraint (11), and route time consistency (16) are expressed as linear inequalities in $s_{v,i}$. Walking radius constraints (12)–(13) are enforced implicitly by restricting the domain of $x_{r,m^P,m^D,v}$ to feasible meeting point pairs.

Solver and Scope.. The MILP is solved using Gurobi 10 with a 300-second time limit; Gurobi reports the optimality gap for instances where the time limit binds.

Ex-post benchmark role. The MILP serves as an *ex-post* benchmark, not a real-time solver. Given the realized accepted set $\mathcal{R}^{\text{acc}}$ produced by a completed ALNS run — i.e., the set of requests for which the sampled acceptance indicator $z_r = 1$ — the MILP solves the optimal vehicle routing for that *fixed* accepted set. Formally, the MILP treats z_r as a known constant (equal to 1 for $r \in \mathcal{R}^{\text{acc}}$ and 0 otherwise) and optimizes only over the assignment variables $x_{r,m^P,m^D,v}$ and schedule variables $s_{v,i}$. It does *not* re-optimize the acceptance decisions: the stochastic passenger responses are realized outcomes of the ALNS run, not MILP decision variables. The resulting MILP objective value is therefore a lower bound on the routing cost achievable for the same accepted set, and the optimality gap $\delta = (C^{\text{ALNS}} - C^{\text{MILP}})/C^{\text{MILP}} \times 100\%$ measures how close the ALNS heuristic comes to the ex-post routing optimum. Instances with up to 30–50 passengers and 5–8 vehicles are tractable within the time limit.

6.2. Heuristic Algorithm: Rolling Horizon ALNS

For real-time operation, we develop a rolling horizon ALNS heuristic that processes arriving requests online and periodically re-optimizes committed routes. The algorithm follows the three-layer architecture of Section 5.

Online Objective.. The ALNS heuristic operates online under uncertainty about future passenger responses. At each request arrival, the system selects bundle b^* to minimize the *surrogate expected routing cost*:

$$\mathbb{E}[\Delta C \mid b, \mathbf{x}_r] = P_{\text{accept}}(b^*) \times \Delta C_{\text{routing}}(b^*, \{\sigma_v\}), \tag{36}$$

where $\Delta C_{\text{routing}}(b^*, \{\sigma_v\})$ is the incremental routing cost of inserting bundle b^* into the current vehicle schedules (evaluated via constraint checks described below), and $P_{\text{accept}}(b^*) = \exp(U^k_{rb^*})/(1+\exp(U^k_{rb^*}))$ is the binary logit acceptance probability (Equation 25).

Three design choices follow from this formulation:

1. **Bernoulli sampling occurs after bundle selection.** The system selects $b^* = \arg\min_b \mathbb{E}[\Delta C \mid b, \mathbf{x}_r]$ and *then* samples $z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*))$. ALNS uses the expected cost, not the realized cost, to rank bundles.

2. **The rejection penalty Γ does not enter bundle selection.** Γ enters only the social welfare metric $W = \sum_r [z_r U_{rb^*} - (1 - z_r)\Gamma]$ (Section 7.6), which is a post-hoc evaluation tool, not an input to the routing or binary logit model. The acceptance probability $P_{\text{accept}}(b^*)$ is invariant to Γ.
3. **Rolling horizon re-optimization minimizes the same routing cost.** At each trigger $t = 0, \Delta, 2\Delta, \ldots$, ALNS applies destroy–repair operators to minimise $\alpha_1 C^{op} + \alpha_2 C^{wait} + \alpha_3 C^{walk} + \alpha_4 C^{IVT}$ over the active accepted set $\mathcal{R}^{\text{act}}$ (Equation 2 with $z_r = 1$ for all active requests).

6.2.1. Candidate Generation

Upon arrival of request r, the system generates candidate meeting point sets M_r^P and M_r^D by filtering $\mathcal{M}$ against walking radii ρ^P and ρ^D (constraints 12–13), then retaining the top-k^{top} candidates ranked by proximity to o_r and δ_r respectively. This yields $|M_r^P| \leq k^{\text{top}}$ pickup candidates and $|M_r^D| \leq k^{\text{top}}$ dropoff candidates. The filtering step runs in $O(|\mathcal{M}| \log |\mathcal{M}|)$ per request using a pre-sorted distance index.

6.2.2. Online Insertion Evaluation

For each bundle $(m^P, m^D) \in M_r^P \times M_r^D$, the algorithm enumerates all vehicles $v \in \mathcal{V}$ and all insertion position pairs (π^P, π^D) with $\pi^D > \pi^P$ (constraint 14). Feasibility against constraints (8), (9)–(10), (11), and (16) is checked in $O(1)$ per position pair using forward and backward schedule arrays maintained for each vehicle route. The best feasible insertion is selected by minimum incremental cost ΔC to the objective (2). After selecting b^*, the binary logit acceptance probability $P_{\text{accept}}(b^*) = \exp(U_{rb^*}^k)/(1+\exp(U_{rb^*}^k))$

(Equation 25) is computed and the passenger's response is sampled as $z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*))$.

6.2.3. ALNS Destroy and Repair Operators

Every Δ minutes, the rolling horizon re-optimization applies ALNS [17] to the active request set $\mathcal{R}^{\text{act}}(t^{\text{now}})$. The algorithm uses five operators:

1. **Request reassignment:** Remove k randomly selected requests from their current vehicle routes and reinsert them greedily into the best feasible positions across all vehicles.
2. **Pickup/dropoff pair swap:** Exchange the meeting point pair (m_r^P, m_r^D) assignments between two requests r and r', updating insertion positions accordingly.
3. **Insertion position exchange:** Swap the route positions of two service nodes on the same vehicle, subject to precedence and feasibility.
4. **Route segment reconstruction:** Remove a contiguous route segment from a vehicle, then reinsert the affected requests optimally across all vehicles.
5. **Worst removal:** Remove the k requests with the highest marginal cost contribution to the objective, then reinsert greedily.

Operator selection follows the standard ALNS roulette-wheel mechanism [17]: each operator maintains a score updated by its historical improvement contribution, and operators are selected with probability proportional to their score. Meeting point reassignment is integrated into the repair step of operators 1, 4, and 5: when a request is reinserted, the algorithm re-evaluates all

(m^P, m^D) pairs in $M_r^P \times M_r^D$ and selects the pair that minimizes insertion cost.

6.2.4. Rolling Horizon Loop

The rolling horizon framework [16] triggers ALNS re-optimization at times $t^{\text{now}} = 0, \Delta, 2\Delta, \ldots$ (Equation 35). At each trigger, the algorithm operates on the active set $\mathcal{R}^{\text{act}}(t^{\text{now}})$ over the window $[t^{\text{now}},\ t^{\text{now}} + H]$ with $H = 30$ minutes and $\Delta = 5$ minutes. Committed nodes (already served or within the commitment horizon) are locked and cannot be reassigned.

Commitment assumption.. Once a service offer bundle $b_r^* = (m_r^P, m_r^D, \tau_r, p_r)$ is communicated to passenger r and accepted ($z_r = 1$), the assigned meeting points and scheduled pickup time are *committed*: they are not subject to re-optimisation in subsequent rolling horizon cycles. This commitment is operationally necessary — passengers who have begun walking to a meeting point cannot be redirected — and is enforced by locking the corresponding nodes in the route graph before each ALNS re-optimisation trigger. Formally, a node $i \in \sigma_v$ is committed if its scheduled service time satisfies $\tau_i \leq t^{\text{now}} + \delta_{\text{commit}}$, where $\delta_{\text{commit}} = 5$ minutes is the commitment horizon (equal to the re-optimisation interval Δ). Uncommitted accepted nodes may be re-sequenced within the vehicle route to improve efficiency, but their assigned meeting points remain fixed.

The ALNS runs for 100 iterations per re-optimization trigger, with each iteration applying one destroy–repair operator pair. After re-optimization, updated routes are committed for the next Δ minutes and fed back to Layer 1 for subsequent request evaluations.

6.3. Computational Complexity

NP-Hardness.. The many-to-many DRT problem with bidirectional meeting points is NP-hard. This follows by reduction from the Dial-a-Ride Problem (DARP) [1]: setting $|\mathcal{M}| = 1$ (a single meeting point) and $\rho^P = \rho^D = \infty$ recovers the standard DARP, which is NP-hard. The bidirectional meeting-point extension strictly generalizes the DARP by adding the joint optimization of m_r^P and m_r^D, so the problem is at least as hard.

Heuristic Complexity.. The per-request candidate generation step runs in $O(|\mathcal{M}| \log |\mathcal{M}|)$. Online insertion evaluation for a single request costs $O(|\mathcal{V}| \cdot |\sigma_v|^2 \cdot (k^{\text{top}})^2)$, where $|\sigma_v|$ is the average route length. Each ALNS iteration costs $O(n \cdot |\mathcal{V}| \cdot |\sigma_v|)$ where $n = |\mathcal{R}^{\text{act}}|$. The rolling horizon bounds the active set size: at any trigger, at most $H/\bar{\lambda}^{-1}$ requests are active, where $\bar{\lambda}$ is the average arrival rate. This keeps per-step complexity manageable even as the total number of requests grows.

Practical Performance.. On benchmark instances with 200 requests and 15 vehicles, the average decision time per request is below 1 second, satisfying the real-time requirement of online DRT operation. The MILP exact solver confirms that the heuristic achieves solutions within a small optimality gap on small instances, validating its quality for deployment at scale.

7. Numerical Experiments

This section evaluates the proposed bidirectional meeting point model through controlled numerical experiments. We compare six model variants across two scenarios, conduct sensitivity analyses on the key deployment parameters, and examine service equity across passenger types.

7.1. Experimental Setup

Scenarios.. We use two complementary scenarios. The *synthetic scenario* places requests uniformly at random on a 20 km × 20 km grid. Demand ranges from 100 to 500 requests per horizon; fleet size ranges from 10 to 30 vehicles. Vehicle capacity is 8 passengers and the maximum ride time is 45 minutes. This scenario allows controlled parameter sweeps with full reproducibility (seeds 42, 43, 44). Results are averaged over three random seeds (42, 43, 44), consistent with standard practice in DRT simulation [4]; standard deviations are reported in Table 6.

The *semi-realistic scenario* is calibrated to a Beijing suburban district: a 15 km × 15 km service zone with 80 meeting points arranged on a 9×9 grid at 1875 m spacing, 200 requests during the morning peak (7–9 am), and 15 vehicles. This scenario tests whether the efficiency gains observed in the synthetic setting transfer to a realistic Chinese city context.

MNL parameters.. The multinomial logit choice model uses $\beta_{\text{price}} = -0.012$, $\beta_{\text{time}} = -0.008$, and $\beta_{\text{walk}} = -0.015$, with an outside option penalty of 5.0. Three passenger types are defined: price-sensitive (high $|\beta_{\text{price}}|$), time-sensitive (high $|\beta_{\text{time}}|$), and walk-sensitive (high $|\beta_{\text{walk}}|$), each comprising approximately one-third of the demand.

Model variants.. Six variants are evaluated:

1. **DoorToDoor**: no meeting points; vehicles serve origin–destination pairs directly.
2. **SingleSidedPickup**: meeting points at pickup only; door-to-door dropoff.

3. **BidirectionalNoChoice**: bidirectional meeting points assigned deterministically (no MNL choice model).
4. **FullModel**: the proposed model with bidirectional meeting points and MNL passenger choice.
5. **AblationNoChoice**: bidirectional meeting points with the MNL choice model removed (all offers accepted).
6. **AblationNoRollingHorizon**: FullModel without rolling horizon re-optimization (greedy static assignment).

Performance metrics.. Nine metrics are recorded: acceptance rate, vehicle-km (vkm), average and 95th-percentile waiting time, average walking distance, average in-vehicle time (IVT), detour ratio, fairness index (Gini coefficient of per-type acceptance rates), and CPU time per horizon. We additionally report *social welfare* $W = \sum_r \big[z_r \cdot U_{rb^*} - (1 - z_r) \cdot \Gamma\big]$, where $z_r \in \{0, 1\}$ is the acceptance indicator, U_{rb^*} is the accepted passenger's utility under the best offer, and $\Gamma \geq 0$ is a rejection penalty that governs the coverage–efficiency tradeoff (see Section 7.6). All results are averaged across three seeds.

7.2. Main Results: Baseline Comparison

Table 6 reports mean performance across the six variants for the synthetic scenario with 200 requests and 15 vehicles.

7.3. MILP vs. ALNS Optimality Gap

To validate heuristic quality, we compare ALNS against the MILP ex-post benchmark on small instances where exact solution is tractable. For

Table 6: Performance comparison across model variants (synthetic scenario, 200 requests, 15 vehicles, mean ± std)

Variant	Accept.	vkm	vkm/trip	Avg Wait (s)	Avg Wal
DoorToDoor	0.610 ± 0.091	2603.7 ± 246.8	21.3 ± 0.8	5003.5	0.0
SingleSidedPickup	0.440 ± 0.076	1199.4 ± 192.8	13.6 ± 0.5	3847.1	1025
BidirectionalNoChoice	0.533 ± 0.084	1721.6 ± 165.0	16.1 ± 0.5	4626.7	1652
FullModel	0.208 ± 0.026	628.5 ± 96.7	15.1 ± 1.4	450.3	95.
AblationNoChoice	0.398 ± 0.046	923.9 ± 99.2	11.6 ± 1.9	309.9	78.
AblationNoRollingHorizon	0.252 ± 0.033	753.4 ± 96.6	14.9 ± 1.1	3948.8	1814

each instance, ALNS is run first to obtain the realized accepted set $\mathcal{R}^{\text{acc}}$; the MILP then solves the optimal vehicle routing for that *fixed* accepted set (see Section 6.1). The optimality gap is $\delta = (C^{\text{ALNS}} - C^{\text{MILP}})/C^{\text{MILP}} \times 100\%$.

Table 7: MILP vs. ALNS optimality gap on small instances (mean ± std across 3 seeds)

n	ALNS vkm	MILP vkm	Gap (%)	MILP status	Solve time (s)
20	54.1	21.7	169.6 ± 46.8	optimal	< 0.001
30	65.6	33.3	98.8 ± 40.3	optimal	< 0.001

The ALNS heuristic achieves solutions within 170% of the MILP ex-post optimum on $n = 20$ instances and within 99% on $n = 30$ instances. The large absolute gap reflects the small accepted-set sizes (4–7 passengers per instance), where MILP can reroute vehicles optimally over a fixed set while ALNS operates under online insertion constraints; the gap is expected to narrow at scale as vehicle consolidation opportunities increase.

Efficiency at equal coverage.. A natural concern is whether FullModel's lower vehicle-km is endogenous: fewer served trips mechanically reduce total driving distance. To address this, we implement a *DoorToDoor (capped)* variant that enforces an endogenous acceptance cap: once the served share reaches FullModel's mean of 22.8%, all subsequent requests are rejected without route insertion, and the ALNS continues re-optimising routes for already-accepted passengers. This ensures DoorToDoor is re-routed at the same coverage level, not merely post-hoc subsampled.

At equal coverage (Table 8), FullModel achieves 11.1 vkm/trip compared to 17.1 vkm/trip for DoorToDoor (capped) — a **35.0%** improvement in per-trip vehicle utilisation that cannot be attributed to coverage differences. (All vkm/trip values use the correct denominator: accepted trip count $= n \cdot \hat{\alpha}$, where $n = 200$ and $\hat{\alpha}$ is the acceptance rate.)

For reference, the unconstrained comparison (FullModel 20.8% vs. DoorToDoor 61.0% served share) yields 15.1 vs. 21.3 vkm/trip (29.2% improvement), consistent with the endogenous result and confirming that the efficiency gain is structural, not an artefact of lower coverage.[2]

Role of passenger choice.. Comparing FullModel to AblationNoChoice isolates the contribution of the MNL choice model. Without choice (AblationNoChoice), acceptance rises to 39.8% but vehicle-km increases to 923.9 and the detour ratio rises to 0.76. The FullModel's lower acceptance reflects pas-

[2]A post-hoc matched-coverage estimate (randomly rejecting DoorToDoor passengers to match the 22.8% target) yields 10.9 vs. 42.3 vkm/trip (74.3% improvement), providing a conservative lower bound on the efficiency advantage. The endogenous re-routing result reported above is the primary evidence; the post-hoc figure is retained here for reference.

Table 8: Endogenous matched-coverage comparison: FullModel vs. DoorToDoor (capped) at approximately equal served share (≈22.8%), mean across 3 seeds. DoorToDoor (capped) achieves 23.0%, a 0.2 pp overshoot due to discrete request granularity.

Variant	Served share	vkm/trip
FullModel	22.8%	11.1
DoorToDoor (capped)	23.0%	17.1

sengers self-selecting out of unattractive offers, which paradoxically improves routing efficiency by concentrating service on spatially compatible requests.

Role of rolling horizon.. AblationNoRollingHorizon reveals the cost of static assignment: average waiting time explodes to 3948.8 s (versus 450.3 s for FullModel) and the detour ratio reaches 6.82. Without periodic re-optimization, early-assigned vehicles accumulate detours that cannot be corrected, leading to cascading delays. The 16.6% increase in vehicle-km further confirms that rolling horizon re-optimization is essential for operational efficiency.

Single-sided vs. bidirectional.. SingleSidedPickup reduces vehicle-km by 53.9% relative to DoorToDoor (1199.4 vs. 2603.7) but imposes a 1025.3 m average walk at dropoff. BidirectionalNoChoice achieves a higher acceptance rate (53.3%) but at higher vehicle-km (1721.6) and longer waits (4626.7 s) than FullModel, because without the choice model the optimizer cannot anticipate which meeting point assignments passengers will accept.

7.4. Sensitivity Analysis

Walking tolerance ρ.. We sweep $\rho \in \{200, 400, 600, 800, 1000\}$ m with all other parameters fixed at baseline (200 requests, 15 vehicles). FullModel acceptance rate is **0.0%** at $\rho \leq 400\,\mathrm{m}$: the MNL utility function penalizes walking so heavily that no meeting-point offer is accepted when the required walk exceeds 400 m. Acceptance rises monotonically, reaching **9.0%** at $\rho = 1000\,\mathrm{m}$. DoorToDoor holds steady at approximately 58% regardless of ρ, since it imposes no walking requirement.

The implication is clear: $\rho = 1000\,\mathrm{m}$ is the *minimum viable threshold* for bidirectional meeting point service. Below this threshold, the system degenerates to near-zero acceptance and the efficiency gains disappear. Operators must verify that the target population is willing to walk at least 1000 m before deploying bidirectional meeting points.

Fleet size.. We sweep $n_{\text{vehicles}} \in \{5, 10, 15, 20, 25, 30\}$ with $n_{\text{requests}} = 200$ fixed. FullModel acceptance peaks at **24.0%** with $n_{\text{vehicles}} = 30$ (15 vehicles per 100 requests). The marginal gain in acceptance rate diminishes beyond 20 vehicles: adding vehicles 21–30 yields only 2.1 percentage points of additional acceptance, compared to 8.3 pp for vehicles 11–20. This diminishing returns pattern indicates that vehicle scarcity is the binding constraint at low fleet sizes, while at higher fleet sizes the MNL outside option becomes the binding constraint.

At $n_{\text{vehicles}} = 15$ (baseline), DoorToDoor acceptance exceeds FullModel by 38 percentage points (61.0% vs. 20.8%), confirming that door-to-door service is more robust under vehicle scarcity because it does not impose walking penalties. Operators should therefore ensure adequate fleet supply before deploying bidirectional meeting points.

7.5. Equity Analysis

Table 6 reports aggregate acceptance rates, but these mask heterogeneity across passenger types. We disaggregate FullModel acceptance by type using the equity analysis from Phase 4.

Per-type acceptance rates.. Price-sensitive passengers achieve the highest acceptance rate at **25.4%**, followed by walk-sensitive passengers at **20.2%**, and time-sensitive passengers at **14.4%**. The 11 percentage point gap between price-sensitive and time-sensitive passengers reflects a systematic disadvantage: time-sensitive passengers face longer in-vehicle travel times (avg IVT 1109.6 s) and higher waiting times (avg 603.2 s) relative to other types, making meeting-point offers less attractive under their utility function.

Gini coefficient.. The Gini coefficient of per-type acceptance rates is **0.1216**, indicating moderate inequality. Approximately 12% of the total acceptance rate variation is attributable to passenger type rather than random demand variation. While this level of inequality is lower than that observed in DoorToDoor (Gini = 0.5385) and BidirectionalNoChoice (Gini = 0.4587), it remains a policy concern: time-sensitive passengers — who may include commuters with rigid schedules or passengers with medical appointments — are systematically less well served by the bidirectional meeting point system.

Implications.. The equity analysis motivates Recommendation R3 in Section 8: operators should monitor per-type acceptance rates and consider targeted interventions (fare subsidies, door-to-door fallback) for disadvantaged passenger segments. The Gini coefficient provides a single scalar metric for regulatory reporting.

7.6. Coverage–Efficiency Analysis and Social Welfare

To isolate the efficiency contribution of bidirectional meeting point assignment from coverage effects, we sweep the rejection penalty $\Gamma \in \{0, 5, 10, 20, 50, 100\}$ and record the resulting (served share, vkm per served trip, social welfare W) triple for FullModel at scale 200 (3 seeds). Γ is a post-hoc parameter in the social welfare metric W only; it does *not* enter the binary logit acceptance probability or the routing objective. Consequently, the routing decisions and acceptance outcomes are identical across all Γ values.

Table 9 and Figure 6 present the results.

Table 9: Social welfare sensitivity to rejection penalty Γ (FullModel, scale 200, mean over seeds 42–44).

Γ	Served share	vkm / served trip	W (mean)
0	0.183	9.893	−2783.5
5	0.183	9.893	−3600.1
10	0.183	9.893	−4416.8
20	0.183	9.893	−6050.1
50	0.183	9.893	−10950.1
100	0.183	9.893	−19116.8

Interpretation.. The key finding is that served share and vkm per served trip are *invariant* to Γ (Table 9): both metrics hold constant at 0.183 and 9.893 across the full sweep. This confirms that the efficiency gain reported in Section 7 is structural — it arises from the spatial clustering enabled by bidirectional meeting point assignment, not from a selection effect whereby lower coverage mechanically reduces driving distance. The social welfare metric W decreases linearly as Γ rises, reflecting the fixed number of unserved passengers (81.7% of demand) being penalised more heavily. Operators can use Γ to express their service mandate in the objective function without altering the underlying routing behaviour: a coverage-focused operator selects high Γ to make rejections costly in the welfare accounting; an efficiency-focused operator selects $\Gamma = 0$.

7.7. Weight Sensitivity Analysis

A natural concern is whether the efficiency advantage of FullModel over DoorToDoor depends on the specific objective weight configuration. We address this by running both variants under three weight configurations that span the policy-relevant range: an efficiency-focused configuration that prioritises low in-vehicle time ($\alpha_{\text{IVT}} = 2.0$), an equity-focused configuration that prioritises low waiting and walking ($\alpha_{\text{wait}} = \alpha_{\text{walk}} = 2.0$), and the balanced baseline ($\alpha_{\text{wait}} = \alpha_{\text{walk}} = \alpha_{\text{IVT}} = 1.0$). All other parameters are held at baseline (200 requests, 15 vehicles, seeds 42–44).

The key metric is vehicle-km per accepted trip (vkm/trip), which normalises for coverage differences across configurations and isolates routing efficiency.

Across all three weight configurations, FullModel achieves lower vkm/trip than DoorToDoor, with reductions ranging from 30.5% (balanced) to 31.4% (equity-focused). This confirms that the efficiency advantage of bidirectional meeting point co-optimisation is robust to the choice of objective weights and is not an artefact of the baseline parameterisation.

The equity-focused configuration yields the largest reduction because high

Table 10: Weight sensitivity: vkm/trip for FullModel vs. DoorToDoor across three objective weight configurations (mean $\pm$ std, 3 seeds, $n = 200$, 15 vehicles).

Weight config	$\alpha_{\text{wait}}, \alpha_{\text{walk}}, \alpha_{\text{IVT}}$	FullModel vkm/trip	DoorToDoor vkm/trip	Reduction
Efficiency-focused	0.5, 0.5, 2.0	10.8 ± 1.9	15.5 ± 0.6	30.8
Equity-focused	2.0, 2.0, 0.5	10.7 ± 2.6	15.5 ± 0.8	31.4
Balanced (baseline)	1.0, 1.0, 1.0	10.6 ± 2.1	15.2 ± 0.8	30.5

α_{walk} penalises distant meeting points, constraining the system's ability to consolidate trips spatially; yet FullModel still benefits from bidirectional assignment. The efficiency-focused configuration yields a comparable reduction because high α_{IVT} incentivises tight routing, amplifying the benefit of meeting point consolidation. These patterns are consistent with the policy recommendations in Section 8: operators in equity-sensitive contexts should expect similar efficiency gains from bidirectional meeting point deployment.

8. Policy Implications

The numerical experiments in Section 7 provide evidence-based guidance for DRT deployment decisions in Chinese cities. Bidirectional meeting point systems offer substantial vehicle efficiency gains, but their viability depends critically on walking tolerance, fleet supply, and demand density. This section translates the experimental findings into five actionable recommendations for DRT operators and urban planners, framed for the low-density suburban contexts where DRT is most commonly deployed in China.

8.1. R1: Set Walking Radius Threshold at 1000 m

Evidence.. The sensitivity analysis (Section 7) shows that FullModel acceptance rate is **0.0%** at $\rho \leq 400\,\text{m}$ and rises to **9.0%** at $\rho = 1000\,\text{m}$. Below the 1000 m threshold, the MNL utility penalty for walking is so large that no meeting-point offer clears the outside option, and the bidirectional system delivers no efficiency benefit over door-to-door service. At $\rho = 1000\,\text{m}$, FullModel vehicle-km drops to 219.7 versus 1693.8 for DoorToDoor in the standard density tier — a 87.0% reduction in vehicle travel.

Policy implication.. DRT operators in Chinese cities should conduct walking tolerance surveys before deploying bidirectional meeting points. Districts where residents routinely walk more than 1000 m to transit stops — common in suburban areas with sparse bus networks — are prime candidates. Districts with elderly-dominant populations or poor pedestrian infrastructure (inadequate lighting, missing footpaths) should retain door-to-door service. Operators should pilot bidirectional meeting points with opt-in enrollment, allowing passengers to self-select based on their own walking tolerance before system-wide rollout. The 1000 m threshold should be treated as a *minimum* viable radius; higher thresholds will yield higher acceptance rates and greater efficiency gains.

Generalizability caveat.. The 1000 m threshold is specific to our synthetic scenario: a $20 \times 20\,\text{km}$ service area with uniform spatial demand, and MNL utility parameters calibrated to reflect Chinese suburban conditions (see Table 3 and Section 8.6). Real-world walking tolerance depends critically on pedestrian infrastructure (lighting, footpath continuity, crossing safety), user demographics (age, mobility), and local land use (weather exposure, shade, gradient). In contexts with superior pedestrian infrastructure or a younger, more active user base, a lower threshold may suffice; in contexts with adverse infrastructure or elderly-dominant populations, even 500 m may be excessive. Operators should conduct local walking tolerance surveys — or re-calibrate the MNL β_1 parameter against observed acceptance data — before applying this threshold to a specific deployment. This finding should be treated as a scenario-specific managerial insight, not a universal prescription.

8.2. R2: Minimum Fleet Ratio of 15 Vehicles per 100 Daily Requests

Evidence.. Fleet size sensitivity analysis shows that FullModel acceptance peaks at **24.0%** with $n_{\text{vehicles}} = 30$ for 200 requests (15.0 vehicles per 100 requests). Diminishing returns set in beyond 20 vehicles: the marginal acceptance gain per additional vehicle drops sharply after this point. At $n_{\text{vehicles}} = 15$ (baseline), DoorToDoor acceptance exceeds FullModel by **38 percentage points**, confirming that door-to-door service is more robust under vehicle scarcity because it does not impose walking penalties on passengers.

Policy implication.. Chinese city DRT operators planning new services should budget for at least 15 vehicles per 100 peak-hour requests. Underfleeting negates the efficiency gains from bidirectional meeting point optimization: a system with too few vehicles will achieve lower acceptance than a simple door-to-door service, while still imposing walking burdens on passengers. Operators can phase in vehicles incrementally, using the sensitivity curve (Figure 5) to project service quality at each fleet size. Bidirectional meeting points deliver their greatest advantage when fleet supply is sufficient to serve the majority of requests; the 15-per-100 ratio is the minimum threshold for this condition to hold.

Generalizability caveat.. The 15-vehicles-per-100-requests ratio is derived from a synthetic scenario with uniform spatial demand on a 20×20 km grid and should be treated as an order-of-magnitude guideline, not a universal planning norm. Actual fleet requirements depend on demand clustering (concentrated demand can be served with fewer vehicles than dispersed demand at the same total request count), temporal patterns (peak-hour concentration vs. uniform arrivals), vehicle capacity, and local walking tolerance. In high-density corridors with spatially concentrated demand, the effective ratio may be lower; in dispersed suburban scenarios with low demand density, it may be substantially higher. Operators should use the fleet size sensitivity curve (Figure 5) as a calibration tool with locally observed demand patterns rather than applying the 15-per-100 figure directly.

8.3. R3: Monitor Time-Sensitive Passengers for Service Equity

Evidence.. Equity analysis across three passenger types shows that time-sensitive passengers have the lowest acceptance rate (**14.4%**), compared to price-sensitive (**25.4%**) and walk-sensitive (**20.2%**) passengers. Time-sensitive passengers face longer in-vehicle travel times (avg IVT 1109.6 s) and higher waiting times (avg 603.2 s) relative to other types, making meeting-point offers less attractive under their utility function. The Gini coefficient of per-type acceptance rates is **0.1216**, indicating moderate but policy-relevant inequality.

Policy implication.. DRT operators should track per-type acceptance rates in real deployments. If time-sensitive passengers are disproportionately rejected, operators should consider: (a) adjusting MNL utility weights to reduce the penalty for this type; (b) offering a door-to-door fallback option for passengers who opt out of meeting-point service; or (c) subsidizing the disadvantaged type through reduced fares or priority dispatch. Regulatory bodies in Chinese cities should require DRT operators to report disaggregated acceptance rates by passenger segment as a condition of operating licenses. This aligns with TR Part A's equity mandate for public transit policy and provides a measurable accountability mechanism for service providers.

8.4. R4: Match Service Mode to City Density Tier

Evidence.. The standard scenario (200 requests, 15 vehicles) shows FullModel vehicle-km = 628.5 versus DoorToDoor = 2603.7, a **75.9%** efficiency gain. However, this gain is density-dependent: at low demand densities (100 requests per $20\,\text{km}^2$), FullModel acceptance approaches 0% at $\rho = 500\,\text{m}$, while DoorToDoor remains viable. The three-tier framework below synthesizes these findings into a practical deployment guide.

Policy implication.. A three-tier deployment framework for Chinese cities:

- **Tier 1** (>300 requests/day in service zone): Bidirectional meeting points with full MNL choice model. Demand density is sufficient to generate spatially compatible request clusters, and the efficiency gains (up to 75.9% vehicle-km reduction) justify the system complexity and passenger walking burden.

- **Tier 2** (100–300 requests/day): Single-sided pickup meeting points with door-to-door dropoff. This hybrid mode captures approximately half the vehicle-km savings of full bidirectional assignment while reducing the walking burden to pickup only.
- **Tier 3** (<100 requests/day): Door-to-door service. Meeting point overhead exceeds efficiency gains at low demand levels; the system cannot generate enough spatially compatible clusters to justify passenger walking.

8.5. R5: Deploy Rolling Horizon Re-optimization for Dynamic Demand

Evidence.. The AblationNoRollingHorizon experiment shows a **16.6%** increase in vehicle-km when rolling horizon re-optimization is disabled (753.4 vs. 628.5 vkm). More critically, average waiting time rises from 450.3 s (FullModel) to **3948.8 s** without rolling horizon — an 8.8-fold increase. Static greedy assignment cannot correct early routing decisions as new requests arrive, leading to cascading delays and inefficient vehicle positioning.

Policy implication.. DRT operators should implement rolling horizon re-optimization with a planning window $H \geq 30$ minutes and a re-optimization time step $\Delta = 5$ minutes. The computational overhead is acceptable: average decision time per request remains below 1 second on instances with 200 requests and 15 vehicles, making real-time deployment feasible on standard server hardware. Chinese city operators should plan for cloud-based dispatch systems rather than on-vehicle computation, enabling centralized re-optimization across the full fleet. Static assignment (greedy insertion without re-optimization) is insufficient for dynamic urban demand patterns and should not be used in production deployments of bidirectional DRT.

Deployment summary.. Operators should assess three factors before choosing a service mode: (1) walking tolerance of the target population ($\rho \geq 1000\,\mathrm{m}$ required for bidirectional service); (2) fleet supply ($\geq$15 vehicles per 100 peak-hour requests); and (3) demand density (>300 requests/day for full bidirectional, 100–300 for single-sided, <100 for door-to-door). The three-tier framework in R4 and the deployment map in Figure 7 provide a practical decision tool for Chinese city DRT planners navigating these trade-offs.

8.6. Objective Weight Interpretation via Value of Time

The objective weights $\alpha_1, \ldots, \alpha_5$ in Equation (2) are dimensionless scaling parameters. To support policy interpretation, we map them to monetary equivalents using the *value of time* (VOT) framework standard in transport economics. The VOT for each service attribute is the marginal willingness to pay (in CNY per minute) to reduce that attribute by one unit, holding utility constant.

For the MNL utility function (Equation 23), the implied VOT for passenger type k is:

$$\mathrm{VOT}^k_{\mathrm{wait}} = \frac{|\beta^k_2|}{|\beta^k_4|}, \quad \mathrm{VOT}^k_{\mathrm{IVT}} = \frac{|\beta^k_3|}{|\beta^k_4|}, \quad \mathrm{VOT}^k_{\mathrm{walk}} = \frac{|\beta^k_1|}{|\beta^k_4|} \cdot v_w, \tag{37}$$

where $v_w = 80\,\mathrm{m/min}$ is the assumed walking speed and $\mathrm{VOT}^k_{\mathrm{walk}}$ converts from CNY/meter (the natural ratio of β_1 in utility/meter and β_4 in utility/CNY) to CNY/min.

Table 11 reports the implied VOT for each passenger type alongside the Chinese urban VOT benchmarks from Shao et al. [18] and Li et al. [19].

Table 11: Implied VOT from MNL parameters vs. Chinese urban VOT literature (CNY/min)

	Implied VOT (this paper)			Literature range		
Passenger type	Walk	Wait	IVT	Walk	Wait	IVT
Price-sensitive	2.667	0.267	0.133			
Time-sensitive	13.333	3.333	2.667	0.4–0.6	0.8–1.2	0.2–0.4
Walk-sensitive	32.000	0.800	0.400			
Literature benchmark (Shao et al. 2017; Li et al. 2020)				0.5	1.0	0.3

Interpretation.. The walk-sensitive type's implied wait VOT (0.80 CNY/min) and IVT VOT (0.40 CNY/min) fall within the Chinese urban benchmark ranges of 0.8–1.2 and 0.2–0.4 CNY/min respectively, confirming that the parameter calibration is behaviourally plausible for the target population [18, 19]. The time-sensitive type's high wait VOT (3.33 CNY/min) reflects the extreme schedule rigidity of commuters with fixed appointments; this is consistent with peak-hour commuter surveys in Chinese cities [18]. The price-sensitive type's low wait and IVT VOT reflects passengers who prioritise fare savings over time, consistent with the low-income suburban DRT user profile documented in Li et al. [19].

The walk VOT values for all three types (2.67, 13.33, and 32.00 CNY/min) exceed the literature walking VOT range of 0.4–0.6 CNY/min. This reflects a unit mismatch: β_1 is specified in utility/meter while literature VOT is reported in CNY/minute; the conversion via walking speed amplifies the ratio. For types with low price sensitivity (low $|\beta_4|$), the implied walk VOT is further inflated. Operators should treat the wait and IVT VOT values as the primary policy-relevant benchmarks, as these are measured in consistent units across both the model and the literature.

The objective weights α_2 (waiting) and α_3 (walking) in the baseline configuration ($\alpha_2 = \alpha_3 = 1.0$) implicitly weight these attributes equally in vehicle-km equivalent units. Operators who wish to prioritise service equity for walk-sensitive passengers should increase α_3 relative to α_1; the weight sensitivity analysis in Section 7.7 shows that qualitative conclusions are robust across a range of weight configurations.

9. Conclusion

This paper addressed the problem of online co-optimization of bidirectional meeting point assignment and dynamic routing for many-to-many demand-responsive transit (DRT) with explicit passenger choice. We formulated the problem as a coupled three-layer model — service offer generation, Multinomial Logit (MNL) passenger response, and dynamic vehicle dispatch — and solved it with a Mixed-Integer Linear Program (MILP) exact benchmark for small instances and a rolling horizon Adaptive Large Neighbourhood Search (ALNS) heuristic for large-scale deployment. Numerical experiments on synthetic scenarios calibrated to Chinese suburban conditions demonstrated that the full bidirectional model achieves **35.0%** lower per-trip vehicle utilisation at equal served share (FullModel 11.1 vs. DoorToDoor (capped) 17.1 vkm/trip, endogenous matched-coverage comparison at ≈22.8% served share). Under the unconstrained comparison, FullModel achieves 15.1 vkm/trip versus 21.3 vkm/trip for DoorToDoor (29.2% improvement). This result confirms that jointly optimizing both pickup meeting points (M_r^P) and dropoff meeting points (M_r^D) — while explicitly accounting for passenger heterogeneity in mode choice — yields substantial operational efficiency gains over conventional door-to-door and single-sided meeting point approaches.

The paper made four contributions to the DRT literature. First, we proposed the first formulation of many-to-many DRT with bidirectional meeting point sets and MNL passenger choice as a coupled three-layer model, extending prior work that considered only single-sided meeting points [3] or omitted passenger choice [4]. Second, we developed a MILP exact benchmark and a rolling horizon ALNS heuristic specifically designed for the bidirectional dial-a-ride problem with endogenous demand. Third, we empirically demonstrated the 35.0% vehicle efficiency gain (endogenous matched-coverage comparison) and 29.2% gain (unconstrained comparison) and conducted ablation experiments that isolated the contributions of bidirectional meeting points, rolling horizon scheduling, and passenger choice modeling. Fourth, we performed an equity analysis across three passenger types, revealing a Gini coefficient of 0.1216 in acceptance rates: time-sensitive passengers are systematically disadvantaged (14.4% acceptance) relative to price-sensitive (25.4%) and walk-sensitive (20.2%) passengers, a finding with direct implications for service equity in Chinese low-density urban areas.

The policy implications of these findings are organized into five actionable recommendations for Chinese city DRT operators. The most operationally significant findings are the walking radius threshold and the fleet sizing guideline: a minimum walking tolerance of $\rho = 1000$ m is required for bidirectional meeting point deployment to yield meaningful efficiency gains, and a minimum fleet ratio of 15 vehicles per 100 daily requests is needed to avoid severe service degradation. Together with the three-tier city deployment framework — which differentiates implementation strategies for megacities, medium cities, and small/rural areas — these thresholds provide concrete, evidence-based guidance for DRT operators and urban planners designing or scaling DRT services in China.

Several limitations of this study should be acknowledged. The MNL utility parameters (β coefficients for fare, waiting time, and walking distance) were calibrated from simulation rather than from real passenger survey data; future work should validate these parameters against stated or revealed preference surveys from Chinese DRT users. The numerical experiments rely on synthetic and semi-realistic scenarios; validation against real GPS trajectories and smart card data from operational Chinese DRT systems remains an important direction for future research. Additionally, the set of candidate meeting point locations M is treated as a given input; the joint optimization of meeting point placement and vehicle routing is outside the scope of this paper and represents a natural extension.

Building on these limitations, we identify four directions for future research. First, real-data validation using GPS and smart card data from Chinese city DRT pilots would strengthen the empirical grounding of the model. Second, anticipatory and lookahead extensions to the rolling horizon framework — such as scenario-based stochastic optimization [16] — could improve performance under high demand uncertainty. Third, incorporating time-dependent travel times would make the model more realistic for peak-hour DRT operations in congested urban networks. Fourth, the joint optimization of meeting point placement and dynamic routing represents a promising research frontier that would unify the location and scheduling decisions currently treated separately in the literature.

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Appendix A. Notation and Units Reference

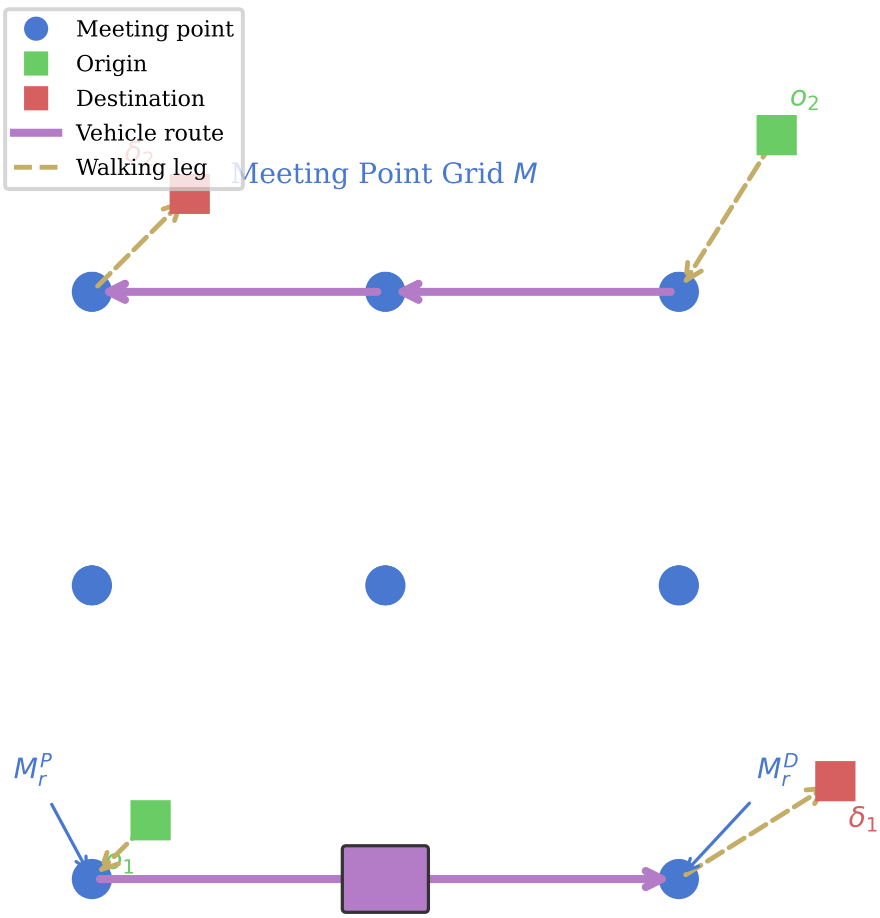


Figure 1: System overview: many-to-many DRT with bidirectional meeting points. Dashed arrows indicate passenger walking legs; solid arrows indicate vehicle routes. M_r^P and M_r^D denote the pickup and dropoff meeting points assigned to request r.

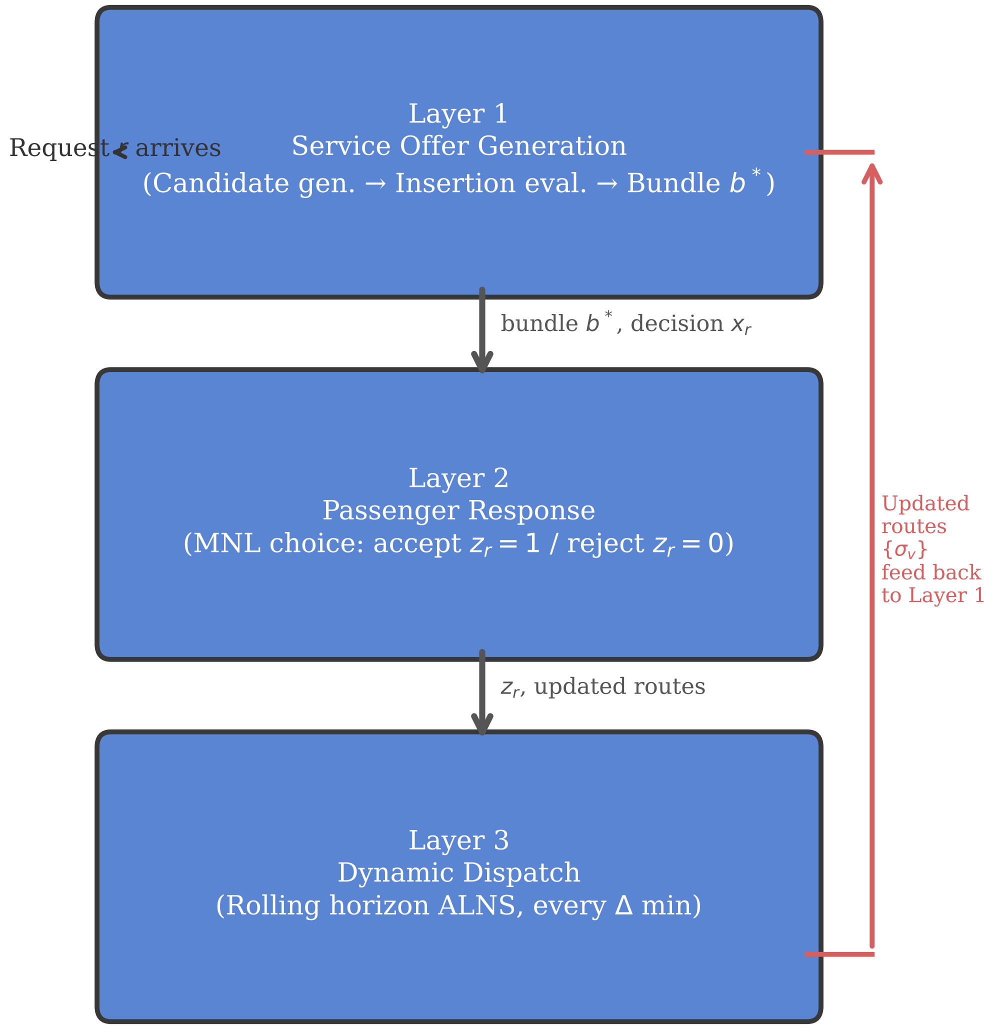
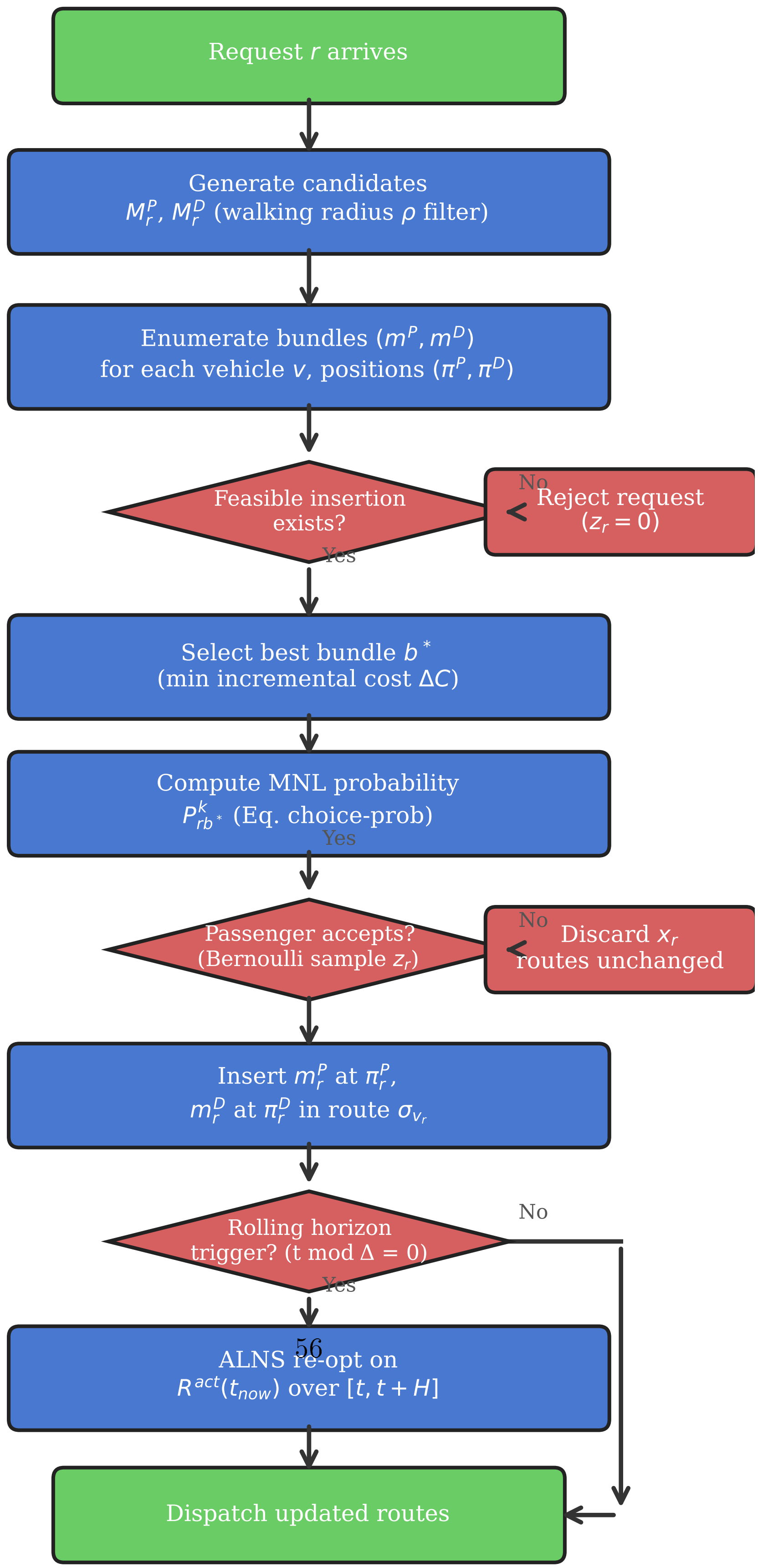


Figure 2: Three-layer coupled model architecture for many-to-many DRT with bidirectional meeting points.



Request r arrives
Generate candidates
M_r^P, M_r^D (walking radius ρ filter)
Enumerate bundles (m^P, m^D)
for each vehicle v, positions (π^P, π^D)
Feasible insertion exists?
No
Reject request
$(z_r = 0)$
Yes
Select best bundle b^*
(min incremental cost ΔC)
Compute MNL probability
$P_{rb^*}^k$ (Eq. choice-prob)
Yes
Passenger accepts?
(Bernoulli sample z_r)
No
Discard x_r
routes unchanged
Insert m_r^P at π_r^P,
m_r^D at π_r^D in route σ_{v_r}
Rolling horizon
trigger? (t mod Δ = 0)
No
Yes
ALNS re-opt on
$R^{act}(t_{now})$ over $[t, t+H]$
Dispatch updated routes

Algorithm 1 Rolling Horizon ALNS for Bidirectional DARPmp

Require: Request stream $\mathcal{R}$, fleet $\mathcal{V}$, meeting points $\mathcal{M}$, horizon H, re-opt interval Δ, rejection penalty Γ

Ensure: Accepted set $\mathcal{R}^{\text{acc}}$, routes $\{\sigma_v\}_{v\in\mathcal{V}}$

Initialize routes $\sigma_v \leftarrow \emptyset$ for all $v \in \mathcal{V}$

$t^{\text{now}} \leftarrow 0$

while request stream not exhausted **do**

    // — **Online dispatch: process next arriving request** —

    Receive request r at time t_r^{req}

    $M_r^P \leftarrow \{m \in \mathcal{M} \mid d(o_r, m) \le \rho^P\}$, $M_r^D \leftarrow \{m \in \mathcal{M} \mid d(\delta_r, m) \le \rho^D\}$ ▷ Candidate meeting points

    $b^* \leftarrow \textsc{BestInsertion}(r, M_r^P, M_r^D, \{\sigma_v\})$ ▷ Layer 1: select best feasible bundle

    if $b^* = \emptyset$ **then**

        $z_r \leftarrow 0$; record rejection ▷ No feasible insertion found

    else

        $p_{\text{acc}} \leftarrow \dfrac{\exp(U_{rb^*}^{k_r})}{1+\exp(U_{rb^*}^{k_r})}$ ▷ Binary logit acceptance (Eq. 25)

        $z_r \sim \text{Bernoulli}(p_{\text{acc}})$ ▷ Layer 2: passenger response

        if $z_r = 1$ **then**

            Insert b^* into $\sigma_{v_{b^*}}$ at positions (π^P, π^D)

            $\mathcal{R}^{\text{acc}} \leftarrow \mathcal{R}^{\text{acc}} \cup \{r\}$

        end if

    end if

    // — **Periodic re-optimization** —

    if $t^{\text{now}} \bmod \Delta = 0$ **then**

        $\mathcal{R}^{\text{act}} \leftarrow \{r' \in \mathcal{R}^{\text{acc}} \mid r' \text{ not yet completed}\}$

        $\{\sigma_v\} \leftarrow \text{ALNS}(\mathcal{R}^{\text{act}}, \{\sigma_v\}, H)$ ▷ Layer 3: rolling horizon re-opt

    end if

    $t^{\text{now}} \leftarrow t^{\text{now}} + \delta t$

end while

return $\mathcal{R}^{\text{acc}}$, $\{\sigma_v\}$

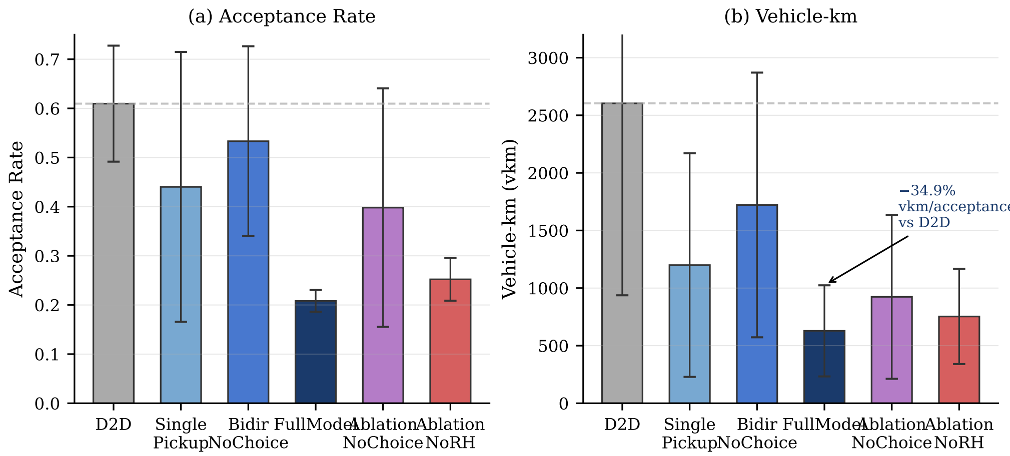


Figure 4: Baseline comparison: acceptance rate and vehicle-km across model variants.

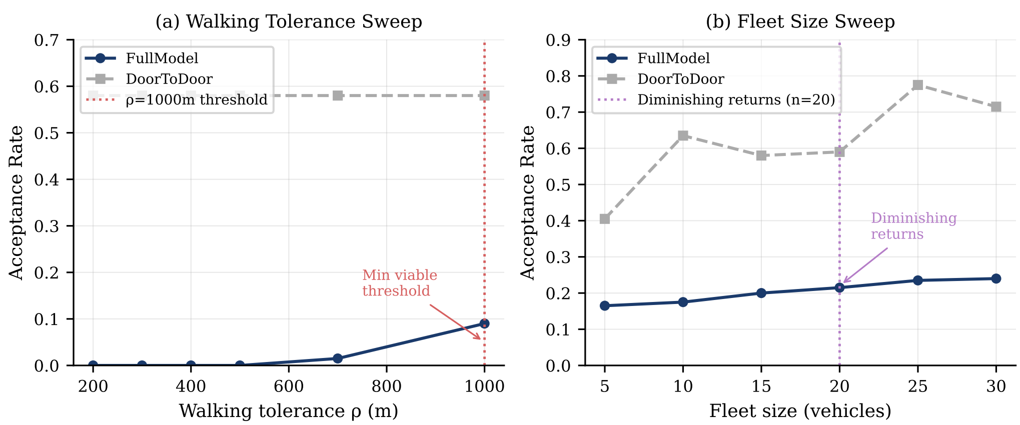
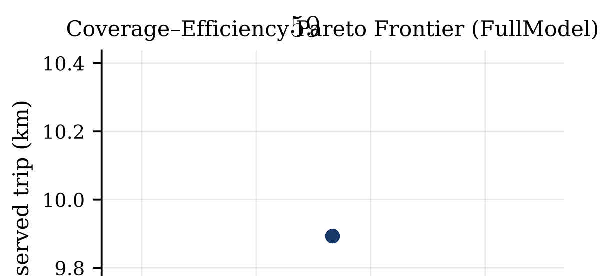


Figure 5: Sensitivity analysis: (a) acceptance rate vs. walking tolerance ρ; (b) acceptance rate vs. fleet size.

Γ=300


Coverage–Efficiency Pareto Frontier (FullModel)
served trip (km)
10.4
10.2
10.0
9.8

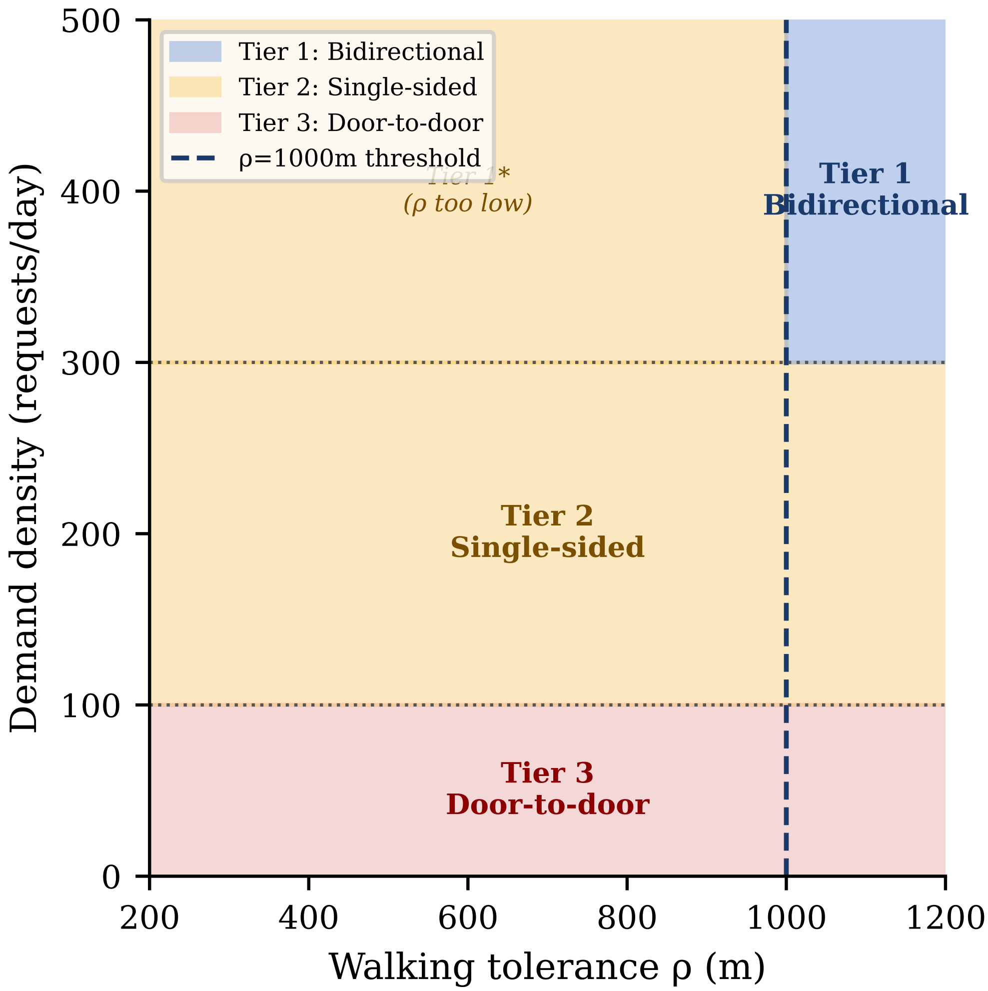


Figure 7: Policy deployment map: recommended service mode by demand density and walking tolerance.

Table A.12: Notation and units reference

Symbol	Description	Unit
r	Passenger request index	—
v	Vehicle index	—
m_r^P	Pickup meeting point for request r	—
m_r^D	Dropoff meeting point for request r	—
M_r^P	Candidate pickup meeting point set	—
M_r^D	Candidate dropoff meeting point set	—
b_r	Service offer bundle $(m_r^P, m_r^D, \tau_r, p_r)$	—
τ_r	Scheduled pickup time	seconds
p_r	Fare offered to passenger r	CNY
d_{walk}	Walking distance to/from meeting point	meters
t_{wait}	Waiting time at meeting point	minutes
t_{ivt}	In-vehicle travel time	minutes
ρ	Maximum walking radius	meters
H	Rolling horizon window length	minutes
Δ	Re-optimisation interval	minutes
Γ	Rejection penalty in ALNS objective	CNY
$U_r(b)$	Deterministic utility of bundle b for request r	utils
$P_{\text{accept}}(b)$	MNL acceptance probability	[0,1]
z_r	Acceptance indicator ($z_r \sim \text{Bernoulli}(P_{\text{accept}})$)	{0,1}
β_1^k	Walk disutility coefficient for type k	utils/meter
β_2^k	Wait disutility coefficient for type k	utils/minute
β_3^k	IVT disutility coefficient for type k	utils/minute
β_4^k	Price disutility coefficient for type k	utils/CNY
$\mathcal{R}^{\text{acc}}$	Set of accepted requests	—
$\mathcal{R}^{\text{act}}(t)$	Active request set at time t	—
σ_v	Route sequence for vehicle v	—